

Clara Dehman

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Department of Physics, University of Alicante, 03690 Alicante, Spain

Education

Ph.D. in Physics

Nov 2019 — Nov 2023

Autonomous University of Barcelona, Spain

- *Thesis Title:* Unveiling the Physics of Neutron Stars: A 3D expedition into MAGneto-Thermal evolution in Isolated Neutron Stars with “MATINS”
- *Honors:* Cum Laude Distinction · International Mention · IEEC Best Thesis Award · SEA Thesis Award (Special Mention)

International Joint M.Sc. in Theoretical Nuclear Physics

Oct 2017 — Sep 2019

Erasmus Mundus NucPhys Program, European Union

Degrees Awarded:

- Laurea Magistrale in Fisica, *University of Padova, Italy* — 110/110 + Cum Laude
- Master de Sciences, Technologies, Santé, Mention Physique, *University of Caen Normandy, France*
- Máster Universitario Erasmus Mundus en Física Nuclear, *Spanish Consortium Universities* — Matrícula de honor

B.Sc. in Physics

Oct 2014 — Aug 2017

Lebanese University, Lebanon — Ranked 1st in Physics Department

Research Positions

Juan de la Cierva Postdoctoral Fellow

Mar 2024 — present

University of Alicante, Spain

- *Project:* Investigate the chiral magnetic effect, magnetic helicity, the origin of magnetar magnetic fields, and long-period transients.
- *Advisor:* Jose A. Pons

Transitional Postdoctoral Researcher

Nov 2023 — Feb 2024

Institute of Space Sciences (ICE-CSIC), Barcelona, Spain

- *Project:* Modeled 3D neutron star magnetic field evolution and compared results with X-ray observations.
- *Advisor:* Nanda Rea

Visiting Ph.D. Fellow

Apr — Dec 2023

Nordita, Stockholm, Sweden

- *Project:* Investigated the role of magnetic helicity in the inverse cascade of neutron star magnetic fields.
- *Advisor:* Axel Brandenburg

Doctoral Researcher

Nov 2019 — Nov 2023

Institute of Space Sciences, Barcelona, Spain

- *Project:* Developed MATINS, a 3D code for the magneto-thermal evolution of isolated neutron stars; applied it to cooling ($\pm$ hyperons), magnetar outbursts, and interior–magnetosphere coupling using physics-informed neural networks.
- *Advisors:* Nanda Rea, Daniele Viganò, Jose A. Pons

Graduate Researcher

Feb — Sep 2019

University of Barcelona, Spain

Grade: 10/10 + Matrícula de Honor

- *Project:* Computed finite-temperature equations of state using the BCPM energy density functional for post-merger neutron stars and late-stage proto-neutron stars.
- *Advisors:* Artur Polls, Xavier Viñas, Mario Centelles

Intern

Sep 2018 — Jan 2019

Laboratory of Corpuscular Physics, Caen, France

- *Project:* Developed metamodeling approaches for nuclear equations of state to interpret experimental data, with a focus on the neutron star mass–radius relation and tidal deformability.
- *Advisors:* Francesca Gulminelli

Intern

Mar — Jun 2018

National Institute for Nuclear Physics, Legnaro Laboratories, Italy

- *Project:* Measured fusion cross sections of a ^{28}Si beam on a ^{100}Mo target at multiple energies to study many-body quantum tunneling.
- *Advisors:* Giovanna Montagnoli, Alberto Stefanini

Fellowships, Grants, & Funding

COMPETITIVE FELLOWSHIPS, GRANTS & SCHOLARSHIPS AS PI

Total secured: €125,500, excluding €92,470 APOSTD Fellowship (declined)

- **May 2026:** *COST Action SCALES*, Coimbra University, Portugal — €1,000
- **Oct 2025:** *International Visitor Grant*, Perimeter Institute, Canada — CAD 2,000
- **Feb 2025 – Jan 2027:** *Juan de la Cierva Postdoctoral Fellowship*, Spanish Ministry of Science and Innovation — nationally competitive (10% success rate) — €74,734
- **Dec 2024:** *IEEC Best Doctoral Thesis Award* — €1,500 awarded to me, plus €1,000 for my thesis supervisors
- **Oct 2024 – Sep 2026:** *APOSTD Postdoctoral Fellowship*, Government of Valencia, Spain — €92,470 (declined in favor of the Juan de la Cierva Fellowship)
- **Apr 2023 – Dec 2023:** *Nordita Visiting PhD Fellowship*, Nordita, Sweden — 80,000 kr
- **Oct 2017 – Sep 2019:** *EACEA Scholarship*, Erasmus Mundus Joint MSc Program, European Education and Culture Executive Agency — €37,000
- **Sep 2012 – Jun 2014:** *English Micro-Scholarship Program*, American Embassy, Lebanon — \$ 2,000

COLLABORATIVE RESEARCH GRANTS AS CONTRIBUTOR

Contributed to research grants totaling €2,944,000

- **Mar 2024 – Dec 2026:** *BLADES (Birth, Life, And Death of Extreme Sources)*, CIPROM 2022/13 Grant, Government of Valencia, Spain — PI: Jose A. Pons, €584,000

- **Mar 2024 – Dec 2026:** *Relativistic Computational Astrophysics*, ASFAE/2022/26 Grant, Government of Valencia, Spain — PI: Miguel A. Aloy, [€300,000](#)
- **Apr 2022 – Dec 2024:** *SGR-Cat 2021 (01269) Grant*, Government of Catalonia, Spain — PI: Vanessa Graber, [€60,000](#)
- **Nov 2019 – Feb 2024:** *Magnesia (The magnetar census and impact of highly magnetic neutron stars on the explosive and transient Universe)* ERC Consolidator Grant, European Commission — PI: Nanda Rea, [€2,000,000](#)

Awards & Certificates

- **Feb 2026:** *Seal of Excellence* — European Commission (Score: 94.6/100)
- **Dec 2024:** *IEEC Best Doctoral Thesis Award 2024*
- **May 2024:** *Special Mention, SEA Thesis Award 2024* — for outstanding research in a highly competitive category
- **Apr 2019:** *Cambridge English: Advanced (CAE)*, Cambridge Assessment English (UK), Grade C (183/210)
- **Mar 2019:** *Certificate in Data Analysis and Machine Learning*, Michigan State University (USA) and University of Oslo (Norway)
- **Apr 2017:** *ERASMUS+ Joint Master Program (NucPhys)* — selected in the top 20% of international applicants by the European Commission

Invited Talks & Seminars (20 Total)

- **Sept 2026:** *Talk*, XII International Conference on Nuclear Physics in Astrophysics, Cluj-Napoca, [Romania](#)
- **Jun 2026:** *Review Talk*, SCALES conference, Superfluid Condensates in Astrophysics and Laboratory Experiments, Coimbra, [Portugal](#)
- **May 2026:** *Talk*, Astrophysical Fluid Dynamics Workshop, Escola d'Enginyeria de Barcelona, [Spain](#)
- **Apr 2026:** *Talk*, SCALES conference, Vortex motion from laboratory to the stars, Warsaw, [Poland](#) [[Link](#)]
- **Apr 2026:** *Strong Gravity Seminar*, Perimeter Institute, Waterloo, [Canada](#)
- **Apr 2026:** *Plasma-Astro Seminar*, Canadian Institute for Theoretical Astrophysics (CITA), Toronto, [Canada](#)
- **Jun 2025:** *Talk*, Fòrum IEEC 2025, Casa Convalescència (UAB), Barcelona, [Spain](#)
- **May 2025:** *Talk*, Extreme Physics of Neutron Star Interiors, Princeton University, [USA](#)
- **May 2025:** *Review Talk*, Workshop on Modern Equation of State and Spectroscopy in Neutron Stars, Alcalá University, [Spain](#)
- **Feb 2025:** *Talk*, Breaking New Ground in Supernova Physics, Fukuoka University, [Japan](#)
- **Dec 2024:** *Review Talk*, IReNA-INT Joint Workshop on Thermal and Magnetic Evolution, University of Washington, [USA](#)
- **Sep 2024:** *Talk*, Frontier Workshop in Astrophysics IV, Palermo, [Italy](#)
- **Jun 2023:** *Astrophysics Seminar*, University of Valencia, [Spain](#)
- **May 2023:** *Astrophysics Seminar*, Nordita, [Sweden](#) [[Link](#)]
- **Jun 2022:** *Astrocoffee Seminar*, Goethe University, [Germany](#) [[Link](#)]
- **May 2022:** *Seminar*, NucPhys Master Program, GANIL, [France](#)

- **Dec 2021:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, [Spain](#) [[Link](#)]
- **Nov 2021:** *Astrophysics Seminar*, University of Alicante, [Spain](#)
- **Dec 2020:** *Talk*, Stars & Compact Objects Meeting, Flatiron Institute, New York, [USA](#)
- **Mar 2019:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, [Spain](#)

Invited Scientific Visit

- **Apr 2026:** *Perimeter Institute for Theoretical Physics*, Ontario, [Canada](#).

Contributed Talks & Seminars (13 Total)

- **Jun 2026:** *Talk*, The X-ray Universe 2026, Elche, [Spain](#)
- **Jun 2026:** *Talk*, NewAthena Rising, ICE-CSIC Barcelona, [Spain](#)
- **Jun 2025:** *Talk*, EAS Annual Meeting, University College Cork, [Ireland](#)
- **Jul 2024:** *Talk*, 10th International Conference on Quarks and Nuclear Physics, University of Barcelona, [Spain](#)
- **Jun 2024:** *Talk*, XMM-Newton Workshop 2024, European Space Agency, Madrid, [Spain](#)
- **May 2024:** *Special Seminar*, Institute of Space Sciences, Barcelona, [Spain](#)
- **Apr 2024:** *Astrophysics Seminar*, University of Alicante, [Spain](#)
- **May 2022:** *Talk*, Pharos Conference, University of La Sapienza, Rome, [Italy](#)
- **Mar 2022:** *Seminar*, Institute of Space Sciences, Barcelona, [Spain](#) [[Link](#)]
- **Nov 2021:** *Talk*, IAU Symposium 363 (online)
- **Jun 2021:** *Talk*, EAS Annual Meeting (online)
- **Nov 2020:** *Seminar*, Institute of Space Sciences, Barcelona, [Spain](#) [[Link](#)]
- **Mar 2020:** *Talk*, PHAROS Conference, Patras, [Greece](#) (canceled due to COVID-19)

Software Development & Open Access Tools

Since Sep 2020: Lead developer of MATINS — a 3D code for the Magneto-Thermal evolution in Isolated Neutron Stars — [MATINS webpage](#)



- *Open access:* github.com/ice-csic-astroexotic/MATINS
- *Lead developers:* **C. Dehman**, D. Viganò, S. Ascenzi
- *Implementation:* Written in Fortran 90 with OpenMP parallelization, optimized for Intel compilers (better performance than GNU).

Supervision

PhD Students — *Co-supervisor; University of Alicante, Spain*

- Joan Llorens Ripoll — Ambipolar diffusion in neutron star cores (since Mar 2025)

MSc Students — *Co-mentor; Institute of Space Sciences, Spain*

- Riccardo Ricci — Magnetar activity and crustal failures (since Oct 2025)
- Marco Ermolli — Impact of microphysics on neutron star cooling (Jan – Jul 2021)

BSc Students — *Main supervisor; University of Alicante, Spain*

- Ahlam Makboul — Glitch timing and magnitude of PSR J0537–6910: probing crust–magnetic field interactions (since Jan 2026)

Teaching (112 Hours Total)

Instructor — Introduction to Modeling in Physics <i>University of Alicante, Spain</i>	Editions 2024 & 2025 <i>45 hr — 95 BSc students</i>
Instructor — Nuclear and Particle Physics <i>University of Alicante, Spain</i>	Editions 2024 & 2025 <i>54 hr — 112 BSc students</i>
Lecturer — Stellar Astrophysics <i>University of Alicante, Spain</i>	Jan — Mar 2025 <i>12 hr — 14 BSc students</i>
Invited Lecturer — Research in Computational Astrophysics <i>University of Alicante, Spain</i>	May 2025 <i>1 hr — 10 MSc students</i>

Scientific Community Service

SCIENTIFIC MEETING ORGANIZATION

SOC: Scientific Organizing Committee; LOC: Local Organizing Committee

- **Sept 2026:** *SOC & LOC, 22nd Pencil Code User Meeting, University of Alicante, Spain* [[Link](#)]
- **Nov 2022:** *LOC, Exploring the Hot and Energetic Universe: the 3rd scientific conference dedicated to the Athena X-ray observatory, Barcelona, Spain* [[Link](#)]
- **May 2022:** *LOC, PHAROS Final Conference 2022: The multi-messenger physics and astrophysics of neutron stars, University of La Sapienza, Rome, Italy* [[Link](#)]

REVIEWING & EVALUATION ACTIVITIES

- **Jan 15, 2026:** *PhD committee member – Juan A. Gil Granados, Univ. of Barcelona, Spain* [[Link](#)]
- **Since Jun 2025:** *Expert evaluator for PhD proposals (2 proposals evaluated), Valencian Agency for Assessment and Prospective, Spain*
- **Since 2022:** *Reviewer for Nature Astronomy; Nature Communications; The Astrophysical Journal; Journal of Physics G; Monthly Notices of the Royal Astronomical Society.*

Outreach

- **Feb 2026:** *Invited outreach talk — International Day of Girls and Women in Science (University of Alicante, Spain)*
- **Jun 2024:** *Invited interview on how scientific discoveries will shape the future of science (University of Elche, Spain)* [[Link](#)]
- **Dec 2023:** *Invited interview on my PhD experience (ICE-CSIC, Barcelona, Spain)* [[Interview](#) | [Newsletter](#)]
- **Nov 2022:** *Astronomical observation — Contributed to organizing a public observation event (Autonomous University of Barcelona, Spain)* [[Link](#)]
- **Sep 2022:** *Research Night — Observation event for school children (Igualada, Spain)*
- **Feb 2022:** *Outreach talk — International Day of Girls and Women in Science (ICE-CSIC, Spain)* [[Link](#)]
- **2021–2023:** *Magnet ICE Outreach — Collaborated with teachers at Gabriel Castellà i Raich to develop science projects promoting student integration in a school facing segregation challenges (Igualada, Spain)*

Press Releases (14 total)

- **Jul 2024:** *Descubren tres estrellas de neutrones jóvenes inusualmente frías para su edad*, [europapress](#)
- **Jul 2024:** *Investigadores de la UA detectan estrellas de neutrones inusualmente frías*, [TODOAlicante](#)
- **Jul 2024:** *Investigadores de la UA participan en el descubrimiento de tres estrellas de neutrones demasiado frías para su edad*, [Actualidad Universitaria](#)
- **Jun 2024:** *NASA's Chandra Peers Into Densest and Weirdest Stars*, [NASA](#)
- **Jun 2024:** *Too young to be so cool: lessons from three neutron stars*, [ESA](#)
- **Jun 2024:** *Un equip d'astrònoms de l'IEEC descobreix tres estrelles de neutrons massa fredes per a la seva edat*, [IEEC](#)
- **Jun 2024:** *Troppo giovani per essere così fredde*, [MEDIA INAF](#)
- **Jun 2024:** *A team of ICE-CSIC astronomers discovers three neutron stars too cold for their age*, [ICE-CSIC](#)
- **Dec 2022:** *Astronomers observe outburst of the young magnetar Swift J1818.0–1607*, [Phys Org](#)
- **Jun 2020:** *NASA discovers youngest 'cosmic baby' neutron star*, [Independent](#)
- **Jun 2020:** *Astronomers just discovered the youngest ever 'baby' dead star*, [CNET](#)
- **Jun 2020:** *Astronomers find baby pulsar – a rare magnetar – born just 240 years ago*, [Astronomy Now](#)
- **Jun 2020:** *Astronomers discover youngest magnetar ever*, [Sci News](#)
- **Jun 2020:** *Hallan estrella 'bebé' que explica origen de las explosiones en el universo*, [La Vanguardia](#)

Memberships, Trainings & Languages

COLLABORATIONS & MEMBERSHIPS

International Astronomical Union (IAU) Junior Member ([since Jul 2026](#)) • SCALES European COST Action CA24139 ([since Nov 2025](#)) • NewAthena Science Community ([since Mar 2025](#)) • Einstein Telescope Collaboration ([since Sep 2022](#)) • AIHUB, Artificial Intelligence Initiative of Spain's National Research Council ([since Mar 2022](#)) • Pharos European COST Action CA16214 (2019 – 2022)

LEADERSHIP TRAINING

Project Citizen (AMIDEAST & American Embassy, Lebanon, [2014](#)) • Intel Youth Innovation Camp (INJAZ, [2013](#)) • Personal Strategic Planning (IAPTL, [2013](#)) • Leadership Program (AMIDEAST & INJAZ, [2012](#))

LANGUAGES

[Arabic](#) (C2, mother tongue) • [English](#) (C2, fluent; Advanced Cambridge 183/210, TOEFL 805/900) • [French](#) (C2, fluent) • [Spanish](#) (B2, intermediate) • [Italian](#) (B2, intermediate)

List of Publications

Curated selection of 23 peer-reviewed articles: 8 as first author, 2 as co-first author (including one in Nature Astronomy), and one invited Living Review, accessible via [ADS](#), [arXiv](#), [ORCID](#), and [Google Scholar](#) — **Citations:** 650+ | **h-index:** 15

SUBMITTED

[23] M. Ronchi, C. Pardo Araujo, V. Graber, N. Rea, **C. Dehman** et al.: 2026. *Connecting radio pulsars, magnetars and XDINSs in a unified evolutionary framework, using simulation-based inference*. Submitted for publication [[PDF](#)]

[22] **C. Dehman**: 2026. *Magnetar field dynamics driven by chiral anomalies without magnetic helicity*. Submitted for publication [[arXiv:2605.08068](#)]

PUBLISHED

[21] J.A. Pons, **C. Dehman**, D. Viganò: 2026. *Magnetic, thermal and rotational evolution of isolated neutron stars*. Living Rev Comput Astrophys 12, 5. [[arXiv:2509.06699](#) | [DOI](#) | cit: 140]

[20] A. Suvorov, **C. Dehman***, J.A. Pons: 2026. *Late-blooming magnetars: awakening as ultra-long period objects after a dormant cooling epoch*. ApJ 1000, 55 (equal contribution by first two authors; CA: C. Dehman) [[arXiv:2505.05373](#) | [DOI](#) | cit: 6]

[19] A. Suvorov, **C. Dehman**, J.A. Pons: 2025. *Revealing the nature of ultra-long period objects with space-based gravitational-wave interferometers*. ApJ 991, 134 [[arXiv:2505.06125](#) | [DOI](#) | cit: 6]

[18] **C. Dehman** and J.A. Pons: 2025. *Magnetar field dynamics shaped by chiral anomalies and helicity*. Phys. Rev. Research 7, 033231 [[arXiv:2505.06196](#) | [DOI](#) | cit: 7]

[17] **C. Dehman** and A. Brandenburg: 2025. *Reality of inverse cascading in neutron star crusts*. A&A 694, A39 [[arXiv:2408.08819](#) | [DOI](#) | cit: 9]

[16] S. Ascenzi, D. Viganò, **C. Dehman** et al.: 2024. *3D code for MAGNETO-Thermal evolution in Isolated Neutron Stars, MATINS: thermal evolution and light curves*. MNRAS 533, 201 [[arXiv:2401.15711](#) | [DOI](#) | cit: 26]

[15] **C. Dehman**, M. Centelles, X. Viñas: 2024. *Impact of the hot inner crust on compact stars at finite temperature*. A&A 687, A236 [[arXiv:2401.16957](#) | [DOI](#) | cit: 7]

[14] A. Marino*, **C. Dehman***, K. Kovlakas*, N. Rea* et al.: 2024. *Constraints on the dense matter equation of state from young and cold isolated neutron stars*. Nature Astron. 8, 1020 (equal contribution by these authors) [[arXiv:2404.05371](#) | [DOI](#) | cit: 32]

[13] J. Urbán, P. Stefanou, **C. Dehman**, J.A. Pons: 2023. *Modeling force-free neutron star magnetospheres using physics-informed neural networks*. MNRAS 524, 32 [[arXiv:2303.11968](#) | [DOI](#) | cit: 20]

[12] **C. Dehman**, D. Viganò, S. Ascenzi et al.: 2023. *3D evolution of neutron star magnetic fields from a realistic core-collapse turbulent topology*. MNRAS 523, 5198 [[arXiv:2305.06342](#) | [DOI](#) | cit: 30]

- [11] **C. Dehman**, J.A. Pons, D. Viganò, N. Rea: 2023. *How bright can old magnetars be? Assessing the impact of magnetized envelopes and field topology on neutron star cooling*. MNRAS Lett. 520, L42 [[arXiv:2301.02261](#)] [DOI](#) | cit: 15]
- [10] A.Y. Ibrahim, A. Borghese, N. Rea et al.: 2023. *Deep X-ray and radio observations of the first outburst of the young magnetar Swift J1818.0-1607*. ApJ 943,20 [[arXiv:2211.12391](#)] [DOI](#) | cit: 16]
- [9] **C. Dehman**, D. Viganò, J.A. Pons, N. Rea: 2023. *3D code for MAGNETO–Thermal evolution in Isolated Neutron Stars, MATINS: the magnetic field formalism*. MNRAS 518, 1222 [[arXiv:2209.12920](#)] [DOI](#) | cit: 39]
- [8] N. Rea, F. Coti Zelati, **C. Dehman** et al.: 2022. *Constraining the nature of the 18 min periodic radio transient GLEAM-X J162759.5-523504.3 via multi-wavelength observations and magneto-thermal simulations*. ApJ 940, 72 [[arXiv:2210.01903](#)] [DOI](#) | cit: 37]
- [7] F. Anzuini, A. Melatos, **C. Dehman** et al.: 2022. *Thermal luminosity degeneracy of magnetized neutron stars with and without hyperon cores*. MNRAS 515, 3014 [[arXiv:2205.14793](#)] [DOI](#) | cit: 11]
- [6] F. Anzuini, A. Melatos, **C. Dehman** et al.: 2022. *Fast cooling and internal heating in hyperon stars*. MNRAS 509, 2609 [[arXiv:2110.14039](#)] [DOI](#) | cit: 28]
- [5] D. Viganò, A. Garcia-Garcia, J.A. Pons, **C. Dehman** et al.: 2021. *Magneto-thermal evolution of neutron stars with coupled Ohmic, Hall and ambipolar effects via accurate finite-volume simulations*. Comput. Phys. Commun. 265, 108001 [[arXiv:2104.08001](#)] [DOI](#) | cit: 61]
- [4] A.M. Stefanini, G. Montagnoli, M. D’Andrea, M. Giacomini, **C. Dehman** et al.: 2021. *New insights into sub-barrier fusion of $^{28}\text{Si} + ^{100}\text{Mo}$* . J. of Phys. G 48, 055101 [[ADS](#)] [DOI](#) | cit: 21]
- [3] F. Coti Zelati, A. Borghese, G.L. Israel et al.: 2021. *The new magnetar SGR J1830-0645 in outburst*. ApJ Lett. 907, L34 [[arXiv:2011.08653](#)] [DOI](#) | cit: 24]
- [2] **C. Dehman**, D. Viganò, N. Rea et al.: 2020. *On the rate of crustal failures in young magnetars*. ApJ Lett. 902, L32 [[arXiv:2010.00617](#)] [DOI](#) | cit: 43]
- [1] P. Esposito, N. Rea, A. Borghese et al.: 2020. *A very young radio-loud magnetar*. ApJ Lett. 896, L30 [[arXiv:2004.04083](#)] [DOI](#) | cit: 79]

MONOGRAPH: PHD THESIS

C. Dehman: 2023. *Unveiling the Physics of Neutron Stars: A 3D expedition into MAGNETO–Thermal evolution in Isolated Neutron Stars with MATINS*, Universitat Autònoma de Barcelona, Catalan Open Research Area [[arXiv:2405.00133](#)] [DOI](#) | cit: 2]